

Jawad Ul Hadi

Senior Backend Engineer · NestJS / Node.js & AI/LLM Integration

Multi-tenant SaaS · APIs · MongoDB · PostgreSQL · GCP · AWS

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PROFESSIONAL SUMMARY

I build backend systems that scale, AI features that ship, and integrations that actually connect. Over the last 7+ years, I've worked across serverless architectures, high-performance databases, and multi-tenant SaaS platforms with strict per-tenant isolation, asynchronous processing, AI integrations with different agents (ChatGPT, Gemini, Anthropic), and deterministic rule-based fallbacks that keep AI features from ever hard-failing. Strong end-to-end expertise in NoSQL/SQL data modelling and API contract design to search, cloud infrastructure, and cost-optimised AI service integration with existing architecture — and equally comfortable owning a data model, a queue worker, or a client conversation.

CORE COMPETENCIES

- **Languages & Frameworks:** TypeScript · Node.js · NestJS · JavaScript · Python · FastAPI · Django & Django REST Framework
- **Architecture & API Design:** Multi-tenant SaaS · Event-Driven Architecture · Modular Monoliths · Domain-Driven Design · REST / GraphQL / WebSocket · Provider Abstraction
- **Data, Search & Performance:** MongoDB / Mongoose · PostgreSQL · MySQL · Redis · MeiliSearch (custom indexing) · Query Optimization · Indexing Strategies · Caching · Full-text & Semantic Search
- **AI & LLM Engineering:** OpenAI SDK · Google Gemini · Anthropic Claude · RAG Pipelines · Structured Output · LLM Observability · Deterministic Rule-Based Fallbacks
- **Async & Event-Driven:** BullMQ + Redis Queues · Idempotent Workers · Webhooks (custom send/receive) · n8n Workflows · HMAC Request Signing
- **Cloud, DevOps & Infrastructure:** GCP · AWS (Lambda, API Gateway) · Docker · CI/CD (GitHub Actions) · Serverless · Zero-Downtime Migrations · Custom Domains & DNS/TLS
- **Integrations:** Stripe (billing) · Twilio (WhatsApp) · Microsoft Graph (Outlook / Teams / Calendar) · Self-hosted Postal SMTP · Google Cloud Storage
- **Security & Governance:** RBAC · JWT / OAuth 2.0 · Tenant Isolation · HMAC Signing · Secrets Management · Audit Logging
- **Document & Data Engineering:** PDF Generation · OCR & Text Extraction (PDF / DOCX / images) · Custom Parsing

SERVICES

- **Backend & API Architecture** — multi-tenant SaaS on NestJS/Node.js; REST, GraphQL & WebSocket contracts.
- **Data & Search Engineering** — MongoDB, PostgreSQL, MySQL, Redis, MeiliSearch; query & index tuning.
- **AI & LLM Integration** — OpenAI, Gemini & Anthropic; RAG, structured output, rule-based fallbacks.
- **Async & Event-Driven Systems** — BullMQ/Redis queues, idempotent workers, webhooks, n8n.

- **Cloud, DevOps & Infrastructure** — GCP & AWS, Docker, CI/CD, zero-downtime migrations, DNS/TLS.
- **Integrations & Security** — Stripe, Twilio, MS Graph, Postal SMTP; RBAC, JWT/OAuth, tenant isolation.

PROFESSIONAL EXPERIENCE

Backend Lead / Senior Backend Engineer

Jan 2024 – Present

MicroAgility Services (Pvt) Ltd. · Islamabad, Pakistan

- Backend lead and architect on a multi-tenant, AI-assisted ATS (private, IP-protected client codebase) — **22 modules, 200+ REST endpoints**, with 3-tier ROOT→PARENT→CHILD tenancy enforced by a verified JWT scope across the entire query layer.
- Built an asynchronous resume-ingestion pipeline on BullMQ/Redis — uploads return immediately while OCR/PDF extraction, AI parsing, indexing, and scoring run in idempotent workers, with deterministic rule-based fallbacks so AI features degrade gracefully and never hard-fail.
- Engineered recruiter email over self-hosted Postal SMTP (nodemailer + Handlebars) with dynamic org branding, plus fully custom send/receive webhook handlers and open/click tracking — no third-party email provider.
- Delivered per-tenant custom career-page domains (DNS/TLS), configurable hiring pipelines, and scorecard engines; built a MeiliSearch sync layer with tenant-scoped indexing for fast full-text candidate and job search.
- Integrated Stripe, Twilio, WhatsApp, and Microsoft Graph (Outlook / Teams / Calendar); enforced tenant isolation at the server level via JWT scopes, eliminating cross-tenant data exposure.
- Kept incremental operational cost per tenant near zero by self-hosting Postal and n8n and leaning on serverless / free-tier services — a deliberate differentiator for cost-sensitive clients.

Backend Software Engineer

Feb 2022 – Jan 2024

MicroAgility Services (Pvt) Ltd. · Islamabad, Pakistan

- Engineered backend services for multi-country HRMS/payroll and project-management platforms — **APAC HRMS**, iAgility, and AgileiBrains — featuring RBAC engines, leave and procurement workflows, and real-time WebSocket collaboration.
- Built APAC HRMS's RBAC engine (9 roles, 31 permission grants), a Leave Bank with Excel import/export, procurement/timesheet/request approval flows, and GCS signed-URL document management.
- Restructured query patterns, indexing, and caching across MySQL/PostgreSQL — cut per-request query volume by **90%+** and reduced dashboard load times from 8-12s to 1-2s.
- Optimised RESTful and GraphQL APIs for ~40% faster data retrieval in production.
- Executed multiple zero-downtime database and server migrations across PostgreSQL, MySQL, and MongoDB, maintaining 99.9% release stability through disciplined Git workflows; containerized services with Docker on GCP and AWS.

Software Developer

Jan 2018 – Mar 2022

Market Icon · Rawalpindi, Pakistan

- Designed serverless AWS Lambda + API Gateway integrations connecting SaaS applications to CRMs and payment gateways, eliminating fixed infrastructure cost while maintaining high availability.
- Built secure, documented APIs with Express.js, FastAPI, and Django REST Framework using JWT and OAuth 2.0 for diverse client projects.

- Developed and maintained full-featured Django applications — ORM data modelling, admin customization, and RESTful exposure via DRF — streamlining internal business workflows.

EDUCATION

B.S. in Computer Science — Government College University, Faisalabad

CERTIFICATIONS

- **Google Cloud** — Generative AI Fundamentals; Cloud Computing Foundations (with IBM).
- **Certified Django Developer** — e-smartdata.org (2025).
- **IBM SkillsBuild** — AI, Cloud Computing, Data, Cybersecurity, IT & Web Development Fundamentals (Credly-verified).

ADDITIONAL INFORMATION

- **Availability:** Open to remote, hybrid, or relocation roles (US / EU / APAC overlap) — full-time or contract.
- **Languages:** English (professional working proficiency) · Urdu (native).